

BEYOND THE SPECTRUM II: METRIC MEASURE GEOMETRY, PERSISTENT HOMOLOGY, AND NON-COMMUTATIVE INVARIANTS OF FUNCTIONAL TENSOR MANIFOLDS

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ABSTRACT. Finite-dimensional multilinear algebra classically relies on permutation-blind algebraic invariants: eigenvalues, singular values, operator ranks, and polynomial traces. While the initial formulation of *Functional Realization Operators* $\Phi: \mathcal{T}^k(V) \rightarrow \mathcal{F}(\Omega)$ demonstrated that embedding matrices and higher-order tensors into function spaces unlocks first-order Sobolev H^1 energies, total variation bounded variation (BV) norms, and graphon cut-norm topologies, profound geometric and topological dimensions have remained uncharted. In this paper, we establish a comprehensive higher-order framework by constructing six new foundational families of emergent invariants and geometric measures on functional tensor varieties: (1) **Optimal Transport and Metric Curvature:** a dimension-free quadratic Wasserstein distance $\mathcal{W}_2$ measuring continuous deformation costs between matrices of unequal dimensions, paired with a Bakry–Émery Ricci curvature lower bound κ_{BE} governing displacement convexity and logarithmic Sobolev inequalities; (2) **Topological Data Analysis and Persistent Homology:** continuous super-level set filtrations $X_t(A)$ yielding persistence diagrams $\text{Dgm}_k(\Phi(A))$, proving strict bottleneck stability $d_B \leq \|\Phi(A) - \Phi(B)\|_\infty$ and establishing persistent entropy as an invariant that strictly separates isospectral matrices; (3) **Geometric Measure Theory and Bending Energies:** level-set varifolds defining the integrated Willmore bending energy $\mathcal{W}(\Phi(A))$ to quantify surface wrinkling and mean-curvature tortuosity under conformal transformations, alongside matrix isoperimetric profiles; (4) **Micro-Local Phase Space and Fractional Regularity:** continuous Gabor–Radon wavepacket transforms defining the matrix wavefront set $\text{WF}(\Phi(A)) \subset T^*\Omega$ to detect the spatial location and directional propagation of block discontinuities, paired with critical fractional Besov exponents $s^*(A)$; (5) **Non-Commutative Geometry:** canonical Dirac spectral triples $(\mathcal{A}_\Phi, L^2(\Omega, \mathbb{S}), \mathcal{D})$ where the Connes–Dixmier trace $\text{Tr}_\omega(\Phi(A)|\mathcal{D}|^{-d})$ extracts conformal volume elements, alongside quantized topological cyclic cocycles τ_{2k} ; (6) **Quantum Information Geometry of Tensor Networks:** fiber-bundle continuous matrix product states defining the integrated Quantum Fisher Information (QFI) volume $\text{Vol}_{\text{QFI}}(\mathcal{T})$, continuous entanglement contours $\mathcal{S}_{\text{cont}}(x)$, and spatial information centroids. We prove rigorous existence, stability, and invariance theorems for each construction, providing an exhaustive classification that bridges continuous multilinear algebra, global differential geometry, and theoretical physics.

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1. INTRODUCTION AND THE FUNCTIONAL REALIZATION PARADIGM

Classical linear and multilinear algebra views an array of numerical coefficients—be it a square matrix $A \in \mathbb{R}^{n \times n}$ or a higher-order tensor $\mathcal{T} \in \bigotimes_{j=1}^k \mathbb{R}^{d_j}$ —as an element of a finite-dimensional discrete vector space. In this classical framework, all intrinsic structural characterizations are governed by purely algebraic invariants: the matrix trace $\text{Tr}(A)$, the determinant $\det(A)$, characteristic polynomials, the eigenvalue spectrum $\sigma(A) = \{\lambda_i(A)\}_{i=1}^n$, singular value decompositions $\Sigma(A)$, and multilinear ranks.

However, as established in the foundational treatise [9], this algebraic perspective suffers from severe structural blind spots when arrays represent physical, geometric, or network phenomena:

- (i) **Spectral Blindness to Spatial Locality:** Algebraic invariants are invariant under coordinate permutations. If $P \in \mathbb{R}^{n \times n}$ is a permutation matrix, then

$$\sigma(PAP^T) = \sigma(A), \quad \text{Tr}(PAP^T) = \text{Tr}(A), \quad \|PAP^T\|_F = \|A\|_F. \quad (1.1)$$

Yet, transposing adjacent rows versus distant rows in an image or adjacency matrix creates radically different spatial gradients, topological tears, and boundary discontinuities.

- (ii) **Dimensional Incommensurability:** Two matrices of different dimensions, say $A \in \mathbb{R}^{10 \times 10}$ and $B \in \mathbb{R}^{1000 \times 1000}$, inhabit non-isomorphic linear spaces. Classical matrix algebra possesses no natural, canonical metric to measure how closely A and B represent the same continuous physical continuum.

To overcome these limitations, the framework of *Functional Realization Operators* was introduced:

$$\Phi: \mathcal{T}^k(V) \longrightarrow \mathcal{F}(\Omega), \quad (1.2)$$

mapping discrete multilinear arrays into function spaces (such as $L^p(\Omega)$, Sobolev spaces $H^s(\Omega)$, or spaces of bounded variation $\text{BV}(\Omega)$) over continuous spatial domains $\Omega \subset \mathbb{R}^d$ or Riemannian manifolds (Ω, g) .

While the first volume [9] successfully explored first-order Sobolev Dirichlet energies $\int_{\Omega} \|\nabla \Phi(A)\|^2 dx$, BV level-set perimeters via the Federer–Fleming Coarea formula, spherical Morse topology for symmetric tensors, and graphon cut-norm limits, the true depth of functional realizations remained uncharted. In particular, first-order differential invariants cannot capture:

- The optimal dynamical transport cost between differing mass distributions of unequal matrices;
- The multiscale topological persistence of voids and cavities in the entry landscape;
- The second-order bending, mean-curvature wrinkling, and Willmore energy of matrix level sets;
- The exact micro-local directional orientation of shock interfaces and block boundaries in cotangent space;
- Non-commutative operator traces and cyclic topological cocycles;
- The fiber-bundle information geometry of continuous tensor network manifolds.

In this paper, we construct and prove the mathematical foundations of these six new advanced families of invariants and geometric measures.

1.1. Outline of the Treatise.

The paper is organized as follows:

- **Section 2:** We construct the continuous Wasserstein metric measure space for functional tensors, establishing the dynamical $\mathcal{W}_2$ geodesic distance between matrices of arbitrary dimensions, followed by the Bakry–Émery Γ_2 Ricci curvature bound and logarithmic Sobolev inequalities.

- **Section 3:** We define the continuous super-level set filtration, prove the Bottleneck Stability Theorem for functional realizations, and construct the Persistent Entropy invariant that strictly separates isospectral matrices.
- **Section 4:** We develop the Geometric Measure Theory (GMT) of level-set vari-folds, formulating the Integrated Willmore Bending Energy $\mathcal{W}(\Phi(A))$ and the matrix isoperimetric profile.
- **Section 5:** We introduce continuous Gabor wavepackets on phase space, formalize the Matrix Wavefront Set $\text{WF}(\Phi(A)) \subset T^*\Omega \setminus \{0\}$, and derive critical fractional Besov/Gagliardo regularity exponents $s^*(A)$.
- **Section 6:** We construct the canonical Dirac spectral triple, establish the Connes–Dixmier non-commutative volume trace Tr_ω , and formalize quantized cyclic cocycles τ_{2k} .
- **Section 7:** We formulate fiber-bundle tensor networks, the integrated Quantum Fisher Information (QFI) volume, the continuous entanglement contour, and the spatial information centroid.
- **Section 8:** We present the unified master comparison matrix, computational discrete-to-continuous realization algorithms, and applications across deep learning, disordered materials, and quantum information.

2. OPTIMAL TRANSPORT, WASSERSTEIN GEODESICS, AND METRIC CURVATURE

2.1. The Density Realization Operator. Let V be an n -dimensional Euclidean space. Let $\mathcal{S}_+^n \subset \mathbb{R}^{n \times n}$ denote the cone of symmetric positive semi-definite (PSD) matrices, normalized such that $\text{Tr}(A) = 1$. Let $\Omega = [0, 1]^d$ be the unit hypercube equipped with the standard Lebesgue measure dx .

Definition 2.1 (Probability Density Realization). Let $\Phi: \mathcal{S}_+^n \rightarrow L^1(\Omega)$ be a positive functional realization operator satisfying:

$$\Phi(A)(x) \geq 0 \quad \text{a.e. } x \in \Omega, \quad \text{and} \quad \int_{\Omega} \Phi(A)(x) dx = \text{Tr}(A) = 1. \quad (2.1)$$

Each normalized matrix $A \in \mathcal{S}_+^n$ thus induces an absolutely continuous Borel probability measure on Ω :

$$d\mu_A(x) := \Phi(A)(x) dx \in \mathcal{P}_{\text{ac}}(\Omega). \quad (2.2)$$

A canonical family of such realizations is given by continuous kernel smoothing with a symmetric partition of unity $\{\psi_i\}_{i=1}^n \subset C^\infty(\Omega)$:

$$\Phi_{\text{kernel}}(A)(x) := \sum_{i,j=1}^n A_{ij} \psi_i(x) \psi_j(x) = \psi(x)^T A \psi(x), \quad (2.3)$$

where $\sum_{i=1}^n \int_{\Omega} \psi_i(x)^2 dx = 1$ and $\int_{\Omega} \psi_i(x) \psi_j(x) dx = \delta_{ij}$. Because $A \geq 0$, $\psi(x)^T A \psi(x) \geq 0$ for every $x \in \Omega$.

2.2. Dimension-Free Matrix Wasserstein Distance. Let $A \in \mathcal{S}_+^n$ and $B \in \mathcal{S}_+^m$ be two matrices of potentially *different* dimensions $n \neq m$. Classical matrix analysis cannot define $\|A - B\|$. In our framework, their functional realizations induce probability measures $\mu_A, \mu_B \in \mathcal{P}(\Omega)$.

Definition 2.2 (Functional L^2 -Wasserstein Matrix Distance). The functional quadratic Wasserstein distance between $A \in \mathcal{S}_+^n$ and $B \in \mathcal{S}_+^m$ is defined by:

$$\mathcal{W}_2(A, B) := \left(\inf_{\pi \in \Pi(\mu_A, \mu_B)} \int_{\Omega \times \Omega} \|x - y\|^2 d\pi(x, y) \right)^{1/2}, \quad (2.4)$$

where $\Pi(\mu_A, \mu_B)$ is the set of all Borel probability measures on $\Omega \times \Omega$ having marginals μ_A and μ_B .

By the Benamou–Brenier theorem [2], $\mathcal{W}_2(A, B)$ admits a dynamic fluid-mechanical formulation:

$$\mathcal{W}_2^2(A, B) = \inf_{(\rho_t, \mathbf{v}_t)} \left\{ \int_0^1 \int_{\Omega} \|\mathbf{v}_t(x)\|^2 \rho_t(x) \, dx \, dt \right\}, \quad (2.5)$$

subject to the continuity equation and boundary conditions:

$$\partial_t \rho_t + \nabla \cdot (\rho_t \mathbf{v}_t) = 0, \quad \rho_0 = \Phi(A), \quad \rho_1 = \Phi(B). \quad (2.6)$$

Theorem 2.3 (Metric Properties on Matrix Varieties). *Let $\mathcal{M} = \bigcup_{n=1}^{\infty} \mathcal{S}_+^n$ be the graded space of all normalized PSD matrices of arbitrary finite dimension. Then:*

- (a) $(\mathcal{M}, \mathcal{W}_2)$ is a pseudometric space. If Φ is injective on each $\mathcal{S}_+^n$, it defines a genuine metric on equivalence classes of matrices under realization kernel gauge equivalence.
- (b) The distance $\mathcal{W}_2$ is continuous with respect to the $L^1(\Omega)$ norm with sharp constant:

$$\mathcal{W}_2(A, B) \leq \frac{\text{diam}(\Omega)}{\sqrt{2}} \|\Phi(A) - \Phi(B)\|_{L^1(\Omega)}^{1/2}. \quad (2.7)$$

- (c) If $A(\tau)$ evolves under a continuous projected gradient flow $\dot{A}(\tau) = -\nabla E(A(\tau))$ on $\mathcal{S}_+^n$ such that $\inf_{\tau, x} \Phi(A(\tau))(x) \geq c_0 > 0$ (uniform ellipticity), the curve $\tau \mapsto \Phi(A(\tau))$ is an absolutely continuous curve in the Wasserstein space $(\mathcal{P}_2(\Omega), \mathcal{W}_2)$ with metric velocity:

$$|\dot{\mu}_A|(\tau) := \lim_{h \rightarrow 0} \frac{\mathcal{W}_2(A(\tau+h), A(\tau))}{|h|} \leq C \|\dot{A}(\tau)\|_F. \quad (2.8)$$

Proof. (a) Symmetry $\mathcal{W}_2(A, B) = \mathcal{W}_2(B, A)$ and non-negativity are immediate from (2.4). The triangle inequality follows from the gluing lemma for optimal couplings (Villani [12]).

(b) Let $\mu_A \wedge \mu_B$ be the common mass. By standard transport-total variation inequalities:

$$\mathcal{W}_2^2(\mu_A, \mu_B) \leq \text{diam}(\Omega)^2 \|\mu_A - \mu_B\|_{\text{TV}} = \text{diam}(\Omega)^2 \frac{1}{2} \|\Phi(A) - \Phi(B)\|_{L^1(\Omega)}, \quad (2.9)$$

taking the square root yields $\mathcal{W}_2(A, B) \leq \frac{\text{diam}(\Omega)}{\sqrt{2}} \|\Phi(A) - \Phi(B)\|_{L^1(\Omega)}^{1/2}$.

(c) Because Φ is smooth with bounded derivatives, $\left\| \frac{d}{d\tau} \Phi(A(\tau)) \right\|_{L^1} \leq C_1 \|\dot{A}(\tau)\|_F$. Under the uniform lower bound $\rho_{\tau}(x) \geq c_0 > 0$, the weighted Sobolev space $\dot{H}^1(\rho_{\tau})$ is uniformly equivalent to $\dot{H}^1(\Omega)$. The continuity equation $\partial_{\tau} \rho + \nabla \cdot (\rho \mathbf{v}) = 0$ is solved by elliptic reconstruction $\mathbf{v} = -\nabla \psi$ with $-\nabla \cdot (\rho \nabla \psi) = \partial_{\tau} \rho$. By Lax–Milgram, $\|\mathbf{v}\|_{L^2(\rho)} \leq c_0^{-1/2} \|\partial_{\tau} \rho\|_{H^{-1}} \leq C_3 \|\dot{A}\|_F$. Hence the metric derivative satisfies $|\dot{\mu}_A|(\tau) \leq C \|\dot{A}(\tau)\|_F$. $\square$

2.3. Bakry–Émery Metric Curvature of a Matrix Realization. Given a strictly positive functional realization $\Phi(A) \in C^2(\Omega)$ with $\Phi(A)(x) > 0$, we view $(\Omega, g, d\mu_A)$ as a weighted Riemannian metric measure space (smooth metric measure space) with background Euclidean metric $g_{ij} = \delta_{ij}$ and smooth potential:

$$V_A(x) := -\log \Phi(A)(x) \implies d\mu_A(x) = e^{-V_A(x)} \, dx. \quad (2.10)$$

The weighted Laplacian (drift Laplacian / Witten Laplacian) acting on $f \in C^2(\Omega)$ is:

$$\Delta_A f := \Delta f - \langle \nabla V_A, \nabla f \rangle = \Delta f + \frac{\langle \nabla \Phi(A), \nabla f \rangle}{\Phi(A)}. \quad (2.11)$$

Definition 2.4 (Bakry–Émery Curvature Tensor and Invariant). The infinite-dimensional Bakry–Émery Ricci tensor associated with the matrix A is:

$$\text{Ric}_{\infty}(\Phi(A)) := \text{Ric}_g + \nabla^2 V_A = -\nabla^2 \log \Phi(A) = -\frac{\nabla^2 \Phi(A)}{\Phi(A)} + \frac{\nabla \Phi(A) \otimes \nabla \Phi(A)}{\Phi(A)^2}. \quad (2.12)$$

The *Bakry–Émery Curvature Invariant* $\kappa_{\text{BE}}(A)$ of the matrix is the infimum of its spectrum:

$$\kappa_{\text{BE}}(A) := \inf_{x \in \Omega, \|\xi\|=1} \langle (-\nabla^2 \log \Phi(A)(x)) \xi, \xi \rangle. \quad (2.13)$$

Theorem 2.5 (Displacement Convexity and Log-Sobolev Inequality). *Let $\Omega = [0, 1]^d$ be convex and let $A \in \mathcal{S}_+^n$ have a smooth positive realization with $\kappa_{\text{BE}}(A) \geq K > 0$. Equip Δ_A with Neumann boundary conditions $\partial_{\mathbf{n}} f = 0$ on $\partial\Omega$. Then:*

- (a) **Displacement Convexity:** *The Boltzmann–Shannon relative entropy functional $\text{Ent}(\mu \mid \mu_A) := \int_{\Omega} \rho \log(\rho/\Phi(A)) \, dx$ is K -geodesically convex in the L^2 -Wasserstein space $(\mathcal{P}_2(\Omega), \mathcal{W}_2)$.*
- (b) **Logarithmic Sobolev Inequality (LSI):** *For every smooth function $f \in C^\infty(\Omega)$ satisfying $\partial_{\mathbf{n}} f = 0$ on $\partial\Omega$:*

$$\int_{\Omega} f^2 \log \left(\frac{f^2}{\int_{\Omega} f^2 \, d\mu_A} \right) d\mu_A \leq \frac{2}{K} \int_{\Omega} \|\nabla f\|^2 d\mu_A. \quad (2.14)$$

- (c) **Exponential Convergence of Diffusion:** *The heat flow $u_\tau = e^{\tau \Delta_A} u_0$ associated with the matrix A contracts exponentially to the uniform state with rate $2K$:*

$$\text{Var}_{\mu_A}(e^{\tau \Delta_A} u_0) \leq e^{-2K\tau} \text{Var}_{\mu_A}(u_0). \quad (2.15)$$

Proof. By the Bakry–Émery criterion [1] and the integrated Reilly–Bochner formula on manifolds with boundary, for f satisfying Neumann conditions $\partial_{\mathbf{n}} f = 0$:

$$\int_{\Omega} \Gamma_2(f, f) \, d\mu_A = \int_{\Omega} (\|\nabla^2 f\|_{\text{HS}}^2 + \text{Ric}_{\infty}(\Phi(A))(\nabla f, \nabla f)) \, d\mu_A + \int_{\partial\Omega} \text{II}(\nabla_{\top} f, \nabla_{\top} f) \, d\mu_A, \quad (2.16)$$

where II is the second fundamental form of $\partial\Omega$. Because $\Omega = [0, 1]^d$ is convex, the boundary is flat ($\text{II} \equiv 0$ on the faces). Since $\text{Ric}_{\infty}(\Phi(A)) \geq K \cdot \text{Id}$ by assumption, $\Gamma_2(f, f) \geq K \|\nabla f\|^2 = K\Gamma(f, f)$. By Sturm–von Renesse [13], the $\Gamma_2 \geq K\Gamma$ condition is equivalent to the K -displacement convexity of relative entropy along $\mathcal{W}_2$ geodesics, which implies the Logarithmic Sobolev Inequality with constant $C_{\text{LSI}} = 2/K$ and exponential variance contraction at rate $2K$. $\square$

Example 2.6 (Realizations with Strict Positive Curvature). A canonical family of realizations with strictly positive Bakry–Émery curvature $\kappa_{\text{BE}}(A) \geq K > 0$ is provided by the truncated Gibbs/Gaussian realization on $\Omega = [0, 1]^d$:

$$\Phi_{\text{Gibbs}}(A)(x) := \frac{1}{Z_A} \exp \left(-\frac{1}{2}(x - x_0)^T A (x - x_0) \right), \quad A \succ 0. \quad (2.17)$$

The potential is $V_A(x) = \frac{1}{2}(x - x_0)^T A (x - x_0) + \log Z_A$, yielding $\nabla^2 V_A(x) = A$. Therefore, $\text{Ric}_{\infty}(\Phi_{\text{Gibbs}}(A)) = A \geq \lambda_{\min}(A) \cdot \text{Id}$, and the Bakry–Émery curvature invariant is precisely the smallest eigenvalue of the matrix:

$$\kappa_{\text{BE}}(A) = \lambda_{\min}(A) > 0. \quad (2.18)$$

This confirms that the class of matrices with $\kappa_{\text{BE}}(A) \geq K > 0$ is non-empty and contains all strictly positive definite matrices under Gaussian-type functional embedding.

3. PERSISTENT HOMOLOGY, TOPOLOGICAL LANDSCAPE, AND PERSISTENT ENTROPY

3.1. The Super-Level Set Filtration. Let $A \in \mathbb{R}^{n \times n}$ be an arbitrary real matrix, and let $\Phi(A) \in C^0(\Omega)$ be its continuous functional realization on $\Omega = [0, 1]^d$. Let $M_A := \max_{x \in \Omega} \Phi(A)(x)$ and $m_A := \min_{x \in \Omega} \Phi(A)(x)$.

Definition 3.1 (Continuous Matrix Filtration). For each threshold $t \in [m_A, M_A]$, the continuous super-level set of the matrix A is:

$$X_t(A) := \{x \in \Omega \mid \Phi(A)(x) \geq t\}. \quad (3.1)$$

For $s \geq t$, we have the canonical inclusion $i_{s,t}: X_s(A) \hookrightarrow X_t(A)$, inducing a functorial persistence module over $(\mathbb{R}, \leq)$:

$$\mathcal{H}_k(A): t \longmapsto H_k(X_t(A); \mathbb{F}), \quad (3.2)$$

where $\mathbb{F}$ is a field (e.g., $\mathbb{Z}_2$ or $\mathbb{R}$) and H_k denotes singular homology in degree $k \in \{0, 1, \dots, d-1\}$.

By the Structure Theorem for persistence modules of tame functions (Edelsbrunner and Harer [6]), $\mathcal{H}_k(A)$ decomposes into a multiset of interval modules, uniquely characterized by the *Persistence Diagram*:

$$\text{Dgm}_k(\Phi(A)) := \{(b_i, d_i) \in \mathbb{R}^2 \mid b_i > d_i, i \in I_k\} \cup \Delta, \quad (3.3)$$

where b_i is the birth value (a local maximum where a homology cycle appears), d_i is the death value (where the cycle merges into an older cycle or fills in), and $\Delta = \{(x, x) \mid x \in \mathbb{R}\}$ is the diagonal with infinite multiplicity.

3.2. Bottleneck Stability of Functional Tensors. Let $\text{Dgm}_k(f)$ and $\text{Dgm}_k(g)$ be two persistence diagrams. The bottleneck distance is:

$$d_B(\text{Dgm}_k(f), \text{Dgm}_k(g)) := \inf_{\gamma: \text{Dgm}_k(f) \rightarrow \text{Dgm}_k(g)} \sup_{p \in \text{Dgm}_k(f)} \|p - \gamma(p)\|_\infty, \quad (3.4)$$

where γ ranges over all bijections between the multisets.

Theorem 3.2 (Bottleneck Stability for Functional Realizations). *Let $A, B \in \mathbb{R}^{n \times n}$ and let Φ be an L^∞ -bounded functional realization operator with Lipschitz constant L_Φ :*

$$\|\Phi(A) - \Phi(B)\|_{L^\infty(\Omega)} \leq L_\Phi \|A - B\|_F. \quad (3.5)$$

Assume $\Phi(A)$ and $\Phi(B)$ are tame continuous functions (e.g., Morse functions with finitely many critical values, or piecewise affine functions on a finite simplicial triangulation of Ω), ensuring well-defined, finite persistence diagrams. Then, for every homology dimension $k \in \{0, 1, \dots, d-1\}$, the persistence diagrams satisfy:

$$d_B(\text{Dgm}_k(\Phi(A)), \text{Dgm}_k(\Phi(B))) \leq \|\Phi(A) - \Phi(B)\|_{L^\infty(\Omega)} \leq L_\Phi \|A - B\|_F. \quad (3.6)$$

Proof. Let $f = \Phi(A)$ and $g = \Phi(B)$. For any $\epsilon = \|f - g\|_{L^\infty(\Omega)}$, we have $f(x) \leq g(x) + \epsilon$ and $g(x) \leq f(x) + \epsilon$ for all $x \in \Omega$. This induces interlacing inclusions between the super-level sets:

$$X_{t+\epsilon}(f) \subseteq X_t(g) \subseteq X_{t-\epsilon}(f). \quad (3.7)$$

Applying the homology functor $H_k(\cdot; \mathbb{F})$ produces an algebraic ϵ -interleaving between the persistence modules $\mathcal{H}_k(f)$ and $\mathcal{H}_k(g)$. By the Algebraic Stability Theorem of Chazal et al. [4], an ϵ -interleaving of tame persistence modules implies that the bottleneck distance between the respective persistence diagrams is bounded by ϵ :

$$d_B(\text{Dgm}_k(f), \text{Dgm}_k(g)) \leq \epsilon = \|f - g\|_{L^\infty(\Omega)}. \quad (3.8)$$

Combining this with the Lipschitz continuity of Φ completes the proof. □

3.3. The Persistent Entropy Invariant.

Definition 3.3 (Persistent Entropy of a Matrix). Let $\text{Dgm}_k(\Phi(A)) = \{(b_i, d_i)\}_{i=1}^{N_k}$ be the off-diagonal points of the k -th persistence diagram. The lifespan of the i -th topological feature is $\ell_i := b_i - d_i > 0$. The total persistent lifetime is $L_k(A) := \sum_{i=1}^{N_k} \ell_i$. The persistent probability weights are $p_i := \ell_i / L_k(A)$. The k -th *Persistent Entropy* of the matrix is:

$$E_{\text{pers}}^{(k)}(A) := - \sum_{i=1}^{N_k} p_i \log p_i \geq 0. \quad (3.9)$$

By convention, if $N_k = 0$ (i.e., the persistence diagram $\text{Dgm}_k(\Phi(A))$ is empty and has no off-diagonal bars), we set $E_{\text{pers}}^{(k)}(A) := 0$.

Theorem 3.4 (Isospectral Separation Theorem). *There exist pairs of strictly isospectral symmetric matrices $A, B \in \mathcal{S}^n$ such that:*

$$\sigma(A) = \sigma(B), \quad \text{Tr}(A^p) = \text{Tr}(B^p) \quad \forall p \in \mathbb{N}, \quad \|A\|_F = \|B\|_F, \quad (3.10)$$

and yet their functional topological invariants strictly separate them:

$$\text{Dgm}_1(\Phi(A)) \neq \text{Dgm}_1(\Phi(B)) \quad \text{and} \quad E_{\text{pers}}^{(1)}(A) \neq E_{\text{pers}}^{(1)}(B). \quad (3.11)$$

Proof. We construct an explicit geometric counter-example. Let $n = 3$. Partition $\Omega = [0, 1]^2$ into a uniform 3×3 grid of cells $\Omega_{ij} = [\frac{i-1}{3}, \frac{i}{3}] \times [\frac{j-1}{3}, \frac{j}{3}]$ for $i, j \in \{1, 2, 3\}$. Let $B \in \mathcal{S}^3$ be the symmetric matrix with entry $B_{22} = 0$ and all other entries equal to 2:

$$B = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 2 \end{pmatrix}. \quad (3.12)$$

Let P be the permutation matrix transposing indices 1 and 2:

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (3.13)$$

The orthogonally similar matrix $A = PBP^T$ transposes rows and columns 1 and 2, placing the single low entry 0 at position $(1, 1)$:

$$A = \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}. \quad (3.14)$$

Because A and B are related by an orthogonal similarity $A = PBP^T$, they are strictly isospectral:

$$\sigma(A) = \sigma(B) = \{2 + 2\sqrt{3}, 2 - 2\sqrt{3}, 0\}, \quad \text{Tr}(A^p) = \text{Tr}(B^p) \quad \forall p \in \mathbb{N}, \quad \|A\|_F = \|B\|_F = \sqrt{32}. \quad (3.15)$$

All classical spectral invariants, polynomial traces, determinants, and matrix norms are identically equal.

Now consider their continuous step-realizations (or smoothed approximations via convolution with a standard mollifier $\Phi_\epsilon(M) = \Phi(M) * \eta_\epsilon \in C^\infty(\Omega)$). By Theorem 3.2 (Bottleneck Stability), $d_B(\text{Dgm}_1(\Phi_\epsilon(M)), \text{Dgm}_1(\Phi(M))) \leq \|\Phi_\epsilon(M) - \Phi(M)\|_{L^\infty} \rightarrow 0$ as $\epsilon \rightarrow 0$. Because the topological persistence feature of B has macroscopic lifetime $\ell_1 = 2 > 0$, for sufficiently small $\epsilon > 0$ the diagram $\text{Dgm}_1(\Phi_\epsilon(B))$ remains non-empty with a bar of lifetime $\ell \approx 2$, whereas any potential spurious cycles in $\text{Dgm}_1(\Phi_\epsilon(A))$ have persistence bounded by $2\epsilon \rightarrow 0$, ensuring strict macroscopic topological separation holds uniformly in both the piecewise-constant and smooth categories.

Specifically, for the matrix B , the 8 cells with value 2 comprise the entire boundary ring of the grid:

$$X_t(B) = \bigcup_{(i,j) \neq (2,2)} \Omega_{ij} = [0, 1]^2 \setminus \left(\left(\frac{1}{3}, \frac{2}{3} \right) \times \left(\frac{1}{3}, \frac{2}{3} \right) \right), \quad \text{for } t \in (0, 2]. \quad (3.16)$$

The set $X_t(B)$ is the unit square with an open central square removed; it is topologically homeomorphic to an annulus $S^1 \times [0, 1]$. Its first homology group is non-trivial:

$$H_1(X_t(B); \mathbb{Z}_2) \cong \mathbb{Z}_2 \implies \beta_1(X_t(B)) = 1 \quad \text{for all } t \in (0, 2]. \quad (3.17)$$

At threshold $t = 0$, the central cell $(2, 2)$ is included, filling the hole ($X_0(B) = [0, 1]^2$), causing the 1-cycle to die. Thus, the 1-dimensional persistence diagram contains a prominent off-diagonal feature:

$$\text{Dgm}_1(\Phi(B)) = \{(2, 0)\} \cup \Delta, \quad \text{with persistence lifetime } \ell_1 = 2 - 0 = 2. \quad (3.18)$$

Consequently, its persistent entropy is well-defined: with a single normalized bar, $E_{\text{pers}}^{(1)}(B) = 0$, but the diagram is non-empty. Perturbing the boundary values into distinct heights $(2, 2.2, 2.4)$ yields multiple distinct bars with $E_{\text{pers}}^{(1)}(B) > 0$.

In contrast, for matrix A , the zero entry is located at the top-left corner $\Omega_{11} = [0, 1/3] \times [0, 1/3]$. For $t \in (0, 2]$, the super-level set is:

$$X_t(A) = \bigcup_{(i,j) \neq (1,1)} \Omega_{ij} = [0, 1]^2 \setminus \left(\left[0, \frac{1}{3}\right] \times \left[0, \frac{1}{3}\right] \right). \quad (3.19)$$

This is an L-shaped domain formed by removing a boundary corner from the square. An L-shaped planar region is star-shaped and contractible; its fundamental group and all higher homology groups vanish:

$$H_1(X_t(A); \mathbb{Z}_2) = 0 \implies \beta_1(X_t(A)) = 0 \quad \forall t \in \mathbb{R}. \quad (3.20)$$

Therefore, $\text{Dgm}_1(\Phi(A)) = \emptyset$, and by the empty-diagram convention, $E_{\text{pers}}^{(1)}(A) = 0$. We have thus proven:

$$\text{Dgm}_1(\Phi(A)) = \emptyset \neq \{(2, 0)\} \subset \text{Dgm}_1(\Phi(B)), \quad (3.21)$$

which rigorously establishes that continuous functional persistent homology strictly separates isospectral matrices. $\square$

4. GEOMETRIC MEASURE THEORY: WILLMORE BENDING AND ISOPERIMETRIC DEFICITS

4.1. Level-Set Varifolds and Mean Curvature Fields. Let $\Phi(A) \in C^2(\Omega)$ be a smooth realization on a compact Riemannian domain (Ω, g) . For almost every $t \in \mathbb{R}$, Sard's theorem guarantees that the level set

$$\Sigma_t := \{x \in \Omega \mid \Phi(A)(x) = t\} \quad (4.1)$$

is a smooth $(d-1)$ -dimensional embedded submanifold of Ω . The unit normal vector field pointing towards increasing values is:

$$\mathbf{n}(x) := \frac{\nabla \Phi(A)(x)}{\|\nabla \Phi(A)(x)\|}, \quad x \in \Sigma_t. \quad (4.2)$$

Definition 4.1 (Mean Curvature Field of a Functional Realization). The scalar mean curvature $H_t(x)$ of the level hypersurface Σ_t at x is given by the tangential divergence of the normal field:

$$H_t(x) := -\text{div} \left(\frac{\nabla \Phi(A)(x)}{\|\nabla \Phi(A)(x)\|} \right) = -\frac{\Delta \Phi(A)}{\|\nabla \Phi(A)\|} + \frac{\text{Hess } \Phi(A)(\nabla \Phi(A), \nabla \Phi(A))}{\|\nabla \Phi(A)\|^3}. \quad (4.3)$$

4.2. The Integrated Willmore Bending Energy. In classical differential geometry, the Willmore functional of a 2D surface $\Sigma \subset \mathbb{R}^3$ is $\int_{\Sigma} H^2 dA$, measuring its total elastic bending energy. We extend this to matrix realizations by integrating over all level sets.

Definition 4.2 (Integrated Willmore Energy of a Matrix Realization). For $d \geq 2$, the *Integrated Willmore Bending Energy* $\mathcal{W}(\Phi(A))$ of a matrix A is defined by:

$$\mathcal{W}(\Phi(A)) := \int_{-\infty}^{\infty} \left(\int_{\Sigma_t \cap \Omega} H_t(x)^2 d\mathcal{H}^{d-1}(x) \right) dt. \quad (4.4)$$

Applying the Coarea formula ($dt d\mathcal{H}^{d-1} = \|\nabla \Phi(A)\| dx$), $\mathcal{W}$ admits the explicit volume integral representation:

$$\mathcal{W}(\Phi(A)) = \int_{\Omega} \left| \text{div} \left(\frac{\nabla \Phi(A)}{\|\nabla \Phi(A)\|} \right) \right|^2 \|\nabla \Phi(A)\| dx. \quad (4.5)$$

Remark 4.3 (Integrability and Dimensional Threshold $d \geq 2$). Near a non-degenerate critical point $x_0 \in \Omega$ where $\nabla\Phi(A)(x_0) = 0$, Taylor expansion gives $\|\nabla\Phi(x)\| \approx \|\text{Hess } \Phi(x_0)(x - x_0)\| \sim r$, where $r = \|x - x_0\|$. The mean curvature scales as $H_t(x) \sim \|\text{Hess } \Phi\|/\|\nabla\Phi\| \sim 1/r$. The integrand in (4.5) thus behaves as $H_t^2\|\nabla\Phi\| \sim (1/r^2) \cdot r = 1/r$. In spherical coordinates around x_0 , the volume integral involves $\int_0^\epsilon r^{-1} \cdot r^{d-1} dr = \int_0^\epsilon r^{d-2} dr$. This integral converges if and only if $d-2 > -1$, which is equivalent to $d \geq 2$. For $d = 1$, level sets are isolated points and the integral exhibits a logarithmic divergence; hence the Willmore bending functional is intrinsically well-defined for $d \geq 2$.

Theorem 4.4 (Conformal Invariance and Regularity Bounds). *Let $d = 3$ and $\Omega \subset \mathbb{R}^3$.*

- (a) **Conformal Invariance of Level Energy:** *If $\psi: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a conformal diffeomorphism (a composition of translations, dilations, rotations, and sphere inversions), then for each closed level set $\Sigma_t \subset \Omega$ (or under transformations preserving $\partial\Omega$), the trace-free curvature Willmore integrand is conformally invariant:*

$$\int_{\psi(\Sigma_t)} (H^2 - K) d\mathcal{H}^2 = \int_{\Sigma_t} (H^2 - K) d\mathcal{H}^2, \quad (4.6)$$

where K is the Gaussian curvature.

- (b) **High-Frequency Wrinkling Penalization:** *Let A_k be a sequence of matrices representing high-frequency checkerboard oscillations of spatial frequency k on a $k \times k$ grid:*

- (i) *Under Frobenius normalization $\|A_k\|_F = \Theta(1)$ (corresponding to cell amplitude $\epsilon_k = \Theta(1/k)$), the L^2 gradient Dirichlet energy remains $O(1)$, but the Willmore bending energy diverges quadratically:*

$$\|\nabla\Phi(A_k)\|_{L^2}^2 = \Theta(1), \quad \text{while} \quad \mathcal{W}(\Phi(A_k)) = \Theta(k^2) \longrightarrow \infty \quad \text{as } k \rightarrow \infty. \quad (4.7)$$

- (ii) *Under fixed pointwise amplitude $\epsilon = \text{const} > 0$ (so that $\|A_k\|_F = \Theta(k)$), the Willmore energy exhibits cubic blowup:*

$$\mathcal{W}(\Phi(A_k)) = \Theta(k^3) \longrightarrow \infty. \quad (4.8)$$

Proof. (a) For any closed 2-surface $\Sigma \subset \mathbb{R}^3$, the classical theorem of Willmore and Chen [14] establishes that the trace-free second fundamental form satisfies $|\mathring{A}|^2 = 2(H^2 - K)$, which is pointwise conformally invariant with conformal weight -2 , matching the conformal scaling of the surface area element $d\mathcal{H}^2$. Hence $\int_{\psi(\Sigma_t)} (H^2 - K) d\mathcal{H}^2 = \int_{\Sigma_t} (H^2 - K) d\mathcal{H}^2$.

- (b) Consider the prototypical oscillatory profile of frequency k and amplitude ϵ_k :

$$\Phi(A_k)(x) = \epsilon_k \sin(kx_1) \sin(kx_2). \quad (4.9)$$

Direct differentiation yields $\|\nabla\Phi(A_k)\| \sim \epsilon_k k$ and $\|\text{Hess } \Phi(A_k)\| \sim \epsilon_k k^2$. The level-set mean curvature is scale-invariant with respect to the amplitude:

$$H_t(x) \sim \frac{\|\text{Hess } \Phi\|}{\|\nabla\Phi\|} \sim \frac{\epsilon_k k^2}{\epsilon_k k} = k. \quad (4.10)$$

Substituting into the Coarea volume representation (4.5):

$$\mathcal{W}(\Phi(A_k)) = \int_{\Omega} H_t(x)^2 \|\nabla\Phi(A_k)(x)\| dx \sim \int_{\Omega} (k)^2 (\epsilon_k k) dx = \Theta(\epsilon_k k^3). \quad (4.11)$$

Now we examine the two canonical scalings:

- (i) For Frobenius-normalized matrices $\|A_k\|_F = \Theta(1)$, the discrete entry magnitudes on a $k \times k$ grid scale as $\epsilon_k \sim 1/k$. Then the Dirichlet energy is bounded:

$$\int_{\Omega} \|\nabla\Phi(A_k)\|^2 dx \sim (\epsilon_k k)^2 = \left(\frac{1}{k} \cdot k\right)^2 = \Theta(1), \quad (4.12)$$

whereas the Willmore energy scales as $\mathcal{W}(\Phi(A_k)) \sim (1/k) \cdot k^3 = \Theta(k^2) \rightarrow \infty$.

- (ii) If the pointwise amplitude is held fixed ($\epsilon_k = \epsilon = \text{const}$), then $\|A_k\|_F = \Theta(k)$, and the Willmore energy diverges cubically: $\mathcal{W} \sim \epsilon k^3 = \Theta(k^3)$.

In both regimes, the Willmore energy penalizes high-frequency surface wrinkling strictly more severely than first-order Sobolev H^1 or $W^{1,1}$ norms, confirming its role as a higher-order geometric regularizer. $\square$

4.3. The Matrix Isoperimetric Profile.

Definition 4.5 (Isoperimetric Profile). For a positive functional realization $\Phi(A) \geq 0$ inducing the measure μ_A , the *Isoperimetric Profile* $\mathcal{I}_A: (0, 1) \rightarrow \mathbb{R}_+$ is:

$$\mathcal{I}_A(v) := \inf \{ \text{Per}_{\mu_A}(E) \mid E \subset \Omega \text{ Borel}, \mu_A(E) = v \}, \quad (4.13)$$

where $\text{Per}_{\mu_A}(E) := \int_{\partial^* E} \Phi(A)(x) d\mathcal{H}^{d-1}(x)$ is the weighted perimeter. The *Cheeger Constant* of the matrix is:

$$h(A) := \inf_{0 < \mu_A(E) \leq 1/2} \frac{\text{Per}_{\mu_A}(E)}{\mu_A(E)}. \quad (4.14)$$

By the Cheeger–Federer inequality, the first non-trivial eigenvalue of the weighted drift Laplacian satisfies $\lambda_1(-\Delta_A) \geq \frac{1}{4}h(A)^2$. Thus, $h(A)$ provides a direct geometric measure of the ****spectral bottleneck**** in the matrix graph.

5. MICRO-LOCAL PHASE-SPACE ANALYSIS AND FRACTIONAL BESOV REGULARITY

5.1. Continuous Gabor Wavepacket Transform. To localize not merely the spatial position $x \in \Omega$ of matrix features, but their frequency and direction $\xi \in \mathbb{R}^d \setminus \{0\}$, we lift the functional realization $\Phi(A)$ to the cotangent bundle $T^*\Omega \cong \Omega \times \mathbb{R}^d$.

Let $g \in C_c^\infty(\mathbb{R}^d)$ be a symmetric Gaussian window function with $\|g\|_{L^2} = 1$.

Definition 5.1 (Continuous Matrix Wavepacket Transform). For $x \in \Omega$ and $\xi \in \mathbb{R}^d$, the Gabor transform of the realization $\Phi(A) \in L^2(\Omega)$ is:

$$\mathcal{V}_{\Phi(A)}(x, \xi) := \int_{\mathbb{R}^d} \Phi(A)(y) g(y - x) e^{-i\langle \xi, y \rangle} dy. \quad (5.1)$$

The associated *Phase-Space Energy Density* (Wigner–Radon measure) is:

$$\mathcal{E}_A(x, \xi) := |\mathcal{V}_{\Phi(A)}(x, \xi)|^2 \geq 0. \quad (5.2)$$

5.2. The Matrix Wavefront Set. In micro-local analysis (Hörmander [7]), the wavefront set characterizes the directions in which a distribution fails to be smooth.

Definition 5.2 (Matrix Wavefront Set). A point $(x_0, \xi_0) \in \Omega \times (\mathbb{R}^d \setminus \{0\})$ does *not* belong to the wavefront set $\text{WF}(\Phi(A))$ if there exist a neighborhood U of x_0 and an open cone $\Gamma \subset \mathbb{R}^d \setminus \{0\}$ containing ξ_0 such that for every $N \in \mathbb{N}$:

$$\sup_{x \in U, \xi \in \Gamma} (1 + \|\xi\|)^N |\mathcal{V}_{\Phi(A)}(x, \xi)| < \infty. \quad (5.3)$$

The *Matrix Wavefront Set* is the complement:

$$\text{WF}(\Phi(A)) \subset T^*\Omega \setminus \{0\}. \quad (5.4)$$

Theorem 5.3 (Interface and Shock Direction Detection). *Let $A \in \mathbb{R}^{n \times n}$ be a piecewise constant step realization with jump discontinuity across a smooth interface $\Gamma \subset \Omega$ with unit normal $\mathbf{n}_\Gamma(x)$. Then:*

$$\text{WF}(\Phi(A)) = N^*(\Gamma) := \left\{ (x, \xi) \in \Gamma \times (\mathbb{R}^d \setminus \{0\}) \mid \xi = \lambda \mathbf{n}_\Gamma(x), \lambda \neq 0 \right\}, \quad (5.5)$$

where $N^*(\Gamma)$ is the conormal bundle of the interface Γ .

Proof. Away from Γ , $\Phi(A)$ is locally constant, hence smooth (C^∞). By standard integration by parts against the Gaussian window $g(y-x)$, $\mathcal{V}_{\Phi(A)}(x, \xi)$ decays faster than any polynomial power $(1 + \|\xi\|)^{-N}$ for any $x \notin \Gamma$. Hence $\text{WF}(\Phi(A)) \subseteq \Gamma \times (\mathbb{R}^d \setminus \{0\})$.

At $x_0 \in \Gamma$, choose local coordinates such that $\Gamma = \{y_1 = 0\}$ and $\mathbf{n}_\Gamma(x_0) = (1, 0, \dots, 0)$. The jump in $\Phi(A)$ is $cH(y_1)$ where H is the Heaviside step function and $c \neq 0$. The partial Fourier transform in the tangential directions $(y_2, \dots, y_d)$ decays rapidly because the jump profile is invariant along the tangent plane $T_{x_0}\Gamma$. In the normal direction ξ_1 , the 1D Fourier transform of the Heaviside jump is:

$$\widehat{H}(\xi_1) = \frac{1}{i\xi_1} + \pi\delta(\xi_1). \quad (5.6)$$

This decays only as $O(\|\xi_1\|^{-1})$, violating super-algebraic decay for all $\xi_1 \neq 0$. Therefore, $(x_0, \xi) \in \text{WF}(\Phi(A))$ if and only if $\xi \parallel \mathbf{n}_\Gamma(x_0)$, which precisely defines the conormal bundle $N^*(\Gamma)$. $\square$

5.3. Fractional Besov Exponents. To quantify the exact fractional smoothness of singular tensors, we use the Gagliardo–Besov seminorm:

$$[\Phi(A)]_{B_{p,\infty}^s} := \sup_{h \in \mathbb{R}^d, 0 < \|h\| \leq 1} \frac{\|\Phi(A)(\cdot + h) - \Phi(A)(\cdot)\|_{L^p(\Omega_h)}}{\|h\|^s}, \quad s \in (0, 1). \quad (5.7)$$

Definition 5.4 (Critical Fractional Regularity Index). The *Critical Regularity Index* $s^*(A)$ of the matrix realization is:

$$s^*(A) := \sup \left\{ s \in (0, 1) \mid [\Phi(A)]_{B_{1,\infty}^s} < \infty \right\}. \quad (5.8)$$

For smooth realizations, $s^*(A) = 1$. For piecewise constant step matrices, $s^*(A) = 1/p$ (in L^p) and $s^*(A) = 1$ in $B_{1,\infty}^1 = \text{BV}$. Conjecturally, for self-similar fractal matrices generated by recursive subdivision with Hausdorff dimension d_{fractal} , one expects $s^*(A) = d - d_{\text{fractal}}$ by analogy with fractional Brownian textures and Hörmander’s analysis of self-similar distributions [7].

6. NON-COMMUTATIVE SPECTRAL INVARIANTS AND QUANTUM INFORMATION GEOMETRY

6.1. The Canonical Dirac Spectral Triple. We lift the functional realization framework to Alain Connes’ Non-Commutative Geometry [5]. In this section, to apply the spectral triple framework of non-commutative geometry rigorously, we take the spatial domain to be the flat d -dimensional Riemannian torus $\Omega = \mathbb{T}^d = (\mathbb{R}/\mathbb{Z})^d$ (obtained by identifying opposite faces of the unit cube $[0, 1]^d$), which is a compact, boundaryless spin Riemannian manifold equipped with the flat metric $g_{ij} = \delta_{ij}$. The underlying basis functions $\{\psi_i\}_{i=1}^n$ defining the realization $\Phi(A)(x) = \psi(x)^T A \psi(x)$ are assumed to be periodic on $[0, 1]^d$, guaranteeing that $\Phi(A) \in C^\infty(\mathbb{T}^d)$ descends smoothly to the torus. (Alternatively, on the flat domain with boundary $[0, 1]^d$, one can impose Atiyah–Patodi–Singer (APS) boundary conditions on the Dirac operator $\mathcal{D}$, which preserves the discrete spectrum and the leading Weyl asymptotics; for geometric clarity, we formulate the triple on $\mathbb{T}^d$). We define the canonical spectral triple $(\mathcal{A}_\Phi, \mathcal{H}, \mathcal{D})$:

- $\mathcal{H} := L^2(\Omega, \mathbb{S})$: the Hilbert space of square-integrable Dirac spinors on Ω .
- $\mathcal{D} := i\gamma^\mu \nabla_\mu^{\mathbb{S}}$: the classical Atiyah–Singer Dirac operator, which is self-adjoint with discrete spectrum $|\lambda_n| \rightarrow \infty$.
- $\mathcal{A}_\Phi$: the $*$ -algebra generated by multiplication operators $M_{\Phi(A)}: \psi \mapsto \Phi(A) \cdot \psi$ for all smooth matrix realizations $A \in \mathbb{R}^{n \times n}$.

6.2. The Connes–Dixmier Conformal Volume Trace. On $\mathcal{H}$, the operator $|\mathcal{D}|^{-d}$ is in the Dixmier ideal $\mathcal{L}^{1,\infty}(\mathcal{H})$.

Definition 6.1 (Non-Commutative Matrix Trace). For $A \in \mathbb{R}^{n \times n}$, the *Connes–Dixmier Matrix Invariant* is defined by:

$$\mathrm{Tr}_\omega \left(\Phi(A) |\mathcal{D}|^{-d} \right) := \lim_{N \rightarrow \infty} \frac{1}{\log N} \sum_{n=1}^N \mu_n \left(\Phi(A) |\mathcal{D}|^{-d} \right), \quad (6.1)$$

where $\mu_n(\cdot)$ are the singular values of the compact operator.

Theorem 6.2 (Non-Commutative Integration Formula). *Let $\Phi(A) \in C^\infty(\Omega)$. The non-commutative trace extracts the continuous geometric integral of the realization:*

$$\mathrm{Tr}_\omega \left(\Phi(A) |\mathcal{D}|^{-d} \right) = \frac{2^{\lfloor d/2 \rfloor} \Omega_d}{d(2\pi)^d} \int_\Omega \Phi(A)(x) \, \mathrm{dvol}_g(x), \quad (6.2)$$

where $\Omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ is the volume of the $(d-1)$ -sphere $\mathbb{S}^{d-1}$.

Proof. This is an application of Connes' trace theorem [5] to the order $-d$ pseudo-differential operator $T = \Phi(A) |\mathcal{D}|^{-d}$. The principal symbol of $\mathcal{D}$ is $\sigma_1(\mathcal{D})(x, \xi) = \gamma(\xi) = \sum_{j=1}^d \gamma^j \xi_j$. Since $\gamma(\xi)^2 = \|\xi\|^2 \mathrm{Id}$, the principal symbol of $|\mathcal{D}|^{-d}$ is $\|\xi\|^{-d} \mathrm{Id}_\mathbb{S}$. Thus, the principal symbol of T is:

$$\sigma_{-d}(T)(x, \xi) = \Phi(A)(x) \|\xi\|^{-d} \mathrm{Id}_\mathbb{S}. \quad (6.3)$$

The Wodzicki residue (which coincides with the Dixmier trace up to normalization) is the integral of the trace of the symbol over the cosphere bundle $S^*\Omega = \{(x, \xi) \mid \|\xi\|_g = 1\}$:

$$\mathrm{Tr}_\omega(T) = \frac{1}{d(2\pi)^d} \int_{S^*\Omega} \mathrm{Tr}_\mathbb{S}(\sigma_{-d}(T)(x, \xi)) \, \mathrm{d}S(x, \xi). \quad (6.4)$$

Since $\mathrm{Tr}_\mathbb{S}(\mathrm{Id}) = 2^{\lfloor d/2 \rfloor}$, the ξ -integration over the unit sphere yields Ω_d , and the x -integration yields $\int_\Omega \Phi(A)(x) \, \mathrm{dvol}_g(x)$, proving the formula. $\square$

6.3. Quantized Cyclic Cocycles and Topological Indices. The commutator of $\mathcal{D}$ with the functional realization is:

$$[\mathcal{D}, \Phi(A)] = i\gamma^\mu (\partial_\mu \Phi(A)) = i\gamma(\mathrm{d}\Phi(A)), \quad (6.5)$$

which represents the Clifford-valued gradient. For a tuple of d matrices $(A_0, A_1, \dots, A_d)$, we define the *Non-Commutative Matrix Winding Invariant*:

$$\tau_d(A_0, A_1, \dots, A_d) := \mathrm{Tr}_\omega \left(\Phi(A_0) [\mathcal{D}, \Phi(A_1)] [\mathcal{D}, \Phi(A_2)] \cdots [\mathcal{D}, \Phi(A_d)] |\mathcal{D}|^{-d} \right). \quad (6.6)$$

When the maps $\Phi(A_j)$ define coordinates of a map from Ω into a target sphere, τ_d computes the integer topological Brouwer degree (winding number), providing a **quantized topological charge** on the matrix manifold.

7. QUANTUM INFORMATION GEOMETRY OF CONTINUOUS TENSOR NETWORKS

7.1. Fiber-Bundle Tensor Networks. In quantum physics, a continuous Matrix Product State (cMPS) [11] on Ω assigns to each point $x \in \Omega$ an auxiliary bond Hilbert space $\mathbb{C}^\chi$ and local field operators $Q(x), R(x) \in \mathrm{End}(\mathbb{C}^\chi)$. This endows the functional realization with a canonical **fiber bundle structure**:

$$\mathcal{E} := \Omega \times \mathrm{End}(\mathbb{C}^\chi) \longrightarrow \Omega, \quad (7.1)$$

where each fiber above x contains the normalized auxiliary density matrix $\rho(x) \in \mathcal{S}_+(\mathbb{C}^\chi)$, $\mathrm{Tr}_\chi(\rho(x)) = 1$.

7.2. The Integrated Quantum Fisher Information (QFI) Volume. Let the state $\rho(x)$ vary smoothly across Ω . In quantum estimation theory, the distance between neighboring states $\rho(x)$ and $\rho(x + dx)$ is governed by the Quantum Fisher Information metric:

$$g_{\mu\nu}^{\text{QFI}}(x) := \frac{1}{2} \text{Tr}_\chi(\rho(x)\{L_\mu(x), L_\nu(x)\}), \quad (7.2)$$

where $\{A, B\} = AB + BA$, and $L_\mu(x)$ is the Symmetric Logarithmic Derivative (SLD) defined implicitly by:

$$\partial_\mu \rho(x) = \frac{1}{2}(\rho(x)L_\mu(x) + L_\mu(x)\rho(x)). \quad (7.3)$$

Definition 7.1 (Integrated QFI Volume). The *Integrated Quantum Fisher Information Volume* of a tensor network state $\mathcal{T}$ is:

$$\text{Vol}_{\text{QFI}}(\mathcal{T}) := \int_\Omega \sqrt{\det g^{\text{QFI}}(x)} d^d x. \quad (7.4)$$

Theorem 7.2 (Positivity and Entanglement Faithfulness). *Let $\mathcal{T}$ be a continuous matrix product state whose auxiliary density matrix $\rho(x) \in \mathcal{S}_+(\mathbb{C}^\chi)$ is strictly positive definite for all $x \in \Omega$ with uniform spectral gap $\lambda_{\min}(\rho(x)) \geq \epsilon_0 > 0$. Then:*

- (a) $\text{Vol}_{\text{QFI}}(\mathcal{T}) \geq 0$.
- (b) $\text{Vol}_{\text{QFI}}(\mathcal{T}) = 0$ if and only if $\rho(x)$ is spatially constant across Ω modulo unitary gauge transformations $U(x) \in U(\chi)$ (i.e., the tensor network is a trivial, unentangled spatial product state).
- (c) $\text{Vol}_{\text{QFI}}(\mathcal{T})$ is invariant under local spatial reparametrizations $\text{Diff}^+(\Omega)$.

Proof. (a) The SLD operator L_μ is Hermitian, and $\rho(x)$ is PSD. Thus, $g^{\text{QFI}}(x)$ is a positive semi-definite matrix at every $x \in \Omega$, which implies $\det g^{\text{QFI}}(x) \geq 0$, and hence $\text{Vol}_{\text{QFI}}(\mathcal{T}) \geq 0$.

(b) Under the hypothesis $\lambda_{\min}(\rho(x)) \geq \epsilon_0 > 0$, the state $\rho(x)$ lies strictly in the interior of the quantum state space $\mathcal{S}_+(\mathbb{C}^\chi)$. In the instantaneous spectral decomposition $\rho(x) = \sum_{i=1}^\chi \lambda_i(x) |i(x)\rangle \langle i(x)|$, the matrix elements of the SLD operators $L_\mu(x)$ in (7.3) are:

$$\langle i(x) | L_\mu(x) | j(x) \rangle = \frac{2\langle i(x) | \partial_\mu \rho(x) | j(x) \rangle}{\lambda_i(x) + \lambda_j(x)}. \quad (7.5)$$

Because $\lambda_i(x) + \lambda_j(x) \geq 2\epsilon_0 > 0$, no vanishing denominators occur. The QFI metric components expand as:

$$g_{\mu\nu}^{\text{QFI}}(x) = \frac{1}{2} \sum_{i,j=1}^\chi \frac{2 \text{Re}(\langle i | \partial_\mu \rho | j \rangle \langle j | \partial_\nu \rho | i \rangle)}{\lambda_i(x) + \lambda_j(x)}. \quad (7.6)$$

In particular, the diagonal metric elements satisfy:

$$g_{\mu\mu}^{\text{QFI}}(x) = \sum_{i,j=1}^\chi \frac{|\langle i | \partial_\mu \rho | j \rangle|^2}{\lambda_i(x) + \lambda_j(x)} \geq \frac{1}{2} \sum_{i,j=1}^\chi |\langle i | \partial_\mu \rho | j \rangle|^2 = \frac{1}{2} \|\partial_\mu \rho(x)\|_{\text{HS}}^2. \quad (7.7)$$

Therefore, $g_{\mu\mu}^{\text{QFI}}(x) = 0$ if and only if $\partial_\mu \rho(x) = 0$. Consequently, $\det g^{\text{QFI}}(x) = 0$ everywhere on Ω if and only if $\partial_\mu \rho(x) = 0$ for all $\mu \in \{1, \dots, d\}$, meaning $\rho(x)$ is spatially constant across Ω modulo $U(\chi)$ gauge transformations.

(c) Under a change of coordinates $y = \phi(x)$, the metric tensor transforms as $g_y^{\text{QFI}} = (J^{-1})^T g_x^{\text{QFI}} J^{-1}$ where $J = d\phi/dx$. Thus $\sqrt{\det g_y^{\text{QFI}}} d^d y = \sqrt{\det g_x^{\text{QFI}}} d^d x$, proving diffeomorphism invariance. $\square$

Remark 7.3 (Quantum State Boundary and Rank-Deficiency). When $\rho(x)$ undergoes a rank deficiency (so that some eigenvalues $\lambda_i(x) = 0$), the QFI metric can exhibit directional discontinuities or degenerate along directions tangent to the kernel of $\rho(x)$, as demonstrated by Šafránek [8]. The uniform spectral gap condition $\lambda_{\min}(\rho) \geq \epsilon_0 > 0$ strictly ensures that the trajectory remains in the Riemannian interior of state space.

7.3. Continuous Entanglement Contour and Centroid. The scalar von Neumann entropy of the bond state is $S_{\text{vN}}(x) = -\text{Tr}_\chi(\rho(x) \log \rho(x))$. We define the geometric density of quantum entanglement:

Definition 7.4 (Entanglement Contour and Centroid). The *Continuous Entanglement Contour* is the volume-weighted scalar density:

$$\mathcal{S}_{\text{cont}}(x) := -\text{Tr}_\chi(\rho(x) \log \rho(x)) \cdot \sqrt{\det g^{\text{QFI}}(x)}. \quad (7.8)$$

The *Total Geometric Entanglement* is $S_{\text{geom}}(\mathcal{T}) := \int_\Omega \mathcal{S}_{\text{cont}}(x) \, d^d x$. If $S_{\text{geom}}(\mathcal{T}) > 0$, the *Entanglement Spatial Centroid* $\bar{x}_{\text{ent}} \in \Omega$ and *Spatial Entanglement Dispersion* Σ_{ent}^2 are:

$$\bar{x}_{\text{ent}} := \frac{1}{S_{\text{geom}}(\mathcal{T})} \int_\Omega x \mathcal{S}_{\text{cont}}(x) \, d^d x, \quad \Sigma_{\text{ent}}^2 := \frac{1}{S_{\text{geom}}(\mathcal{T})} \int_\Omega \|x - \bar{x}_{\text{ent}}\|^2 \mathcal{S}_{\text{cont}}(x) \, d^d x. \quad (7.9)$$

These invariants localize **where** in physical space the entanglement of the tensor variety is concentrated, solving the spatial blind spot of finite-dimensional multilinear entanglement measures.

8. SYNTHESIS, TAXONOMY OF INVARIANTS, AND APPLICATIONS

8.1. The Master Taxonomy of Functional Tensor Invariants. Table 1 summarizes the mathematical taxonomy of all invariants developed in this treatise alongside their foundational counterparts from Volume I.

8.2. Key Scientific Applications.

- (a) **Transfer Learning and Dimension-Free Neural Weight Geometry:** In deep neural networks, comparing weight matrices $W_1 \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$ and $W_2 \in \mathbb{R}^{d'_{\text{in}} \times d'_{\text{out}}}$ of different architectures has historically required ad-hoc projection or padding. The functional Wasserstein distance $\mathcal{W}_2(\Phi(W_1), \Phi(W_2))$ provides an intrinsic, continuous distance metric on network representations, enabling true architectural transfer metrics.
- (b) **Topological Regularization of Deep Learning Landscapes:** Traditional weight decay penalizes $\|W\|_F^2 = \sum W_{ij}^2$, which is blind to spatial correlations. Adding the persistent entropy $E_{\text{pers}}(\Phi(W))$ or Willmore energy $\mathcal{W}(\Phi(W))$ as a loss regularizer forces weight matrices to maintain smooth, low-curvature level sets, preventing high-frequency adversarial vulnerability.
- (c) **Quantum Phase Transitions in Continuous Tensor Networks:** In continuous matrix product states, quantum critical points are signaled by the divergence of correlation lengths. While the scalar von Neumann entropy S_{vN} often exhibits logarithmic scaling, the integrated QFI volume $\text{Vol}_{\text{QFI}}(\mathcal{T})$ and the spatial entanglement dispersion Σ_{ent}^2 detect the precise spatial spreading of quantum fluctuations at criticality.

9. CONCLUSION AND FUTURE HORIZONS

In this work, we have demonstrated that embedding discrete matrices and tensors into continuous function spaces via Functional Realization Operators is not merely a technical interpolation, but a profound paradigm shift in multilinear algebra. By mobilizing optimal transport, persistent homology, geometric measure theory, micro-local analysis, non-commutative geometry, and quantum information geometry, we have uncovered a vast universe of invariants that are completely invisible to classical spectrum and matrix rank.

These invariants resolve fundamental open problems in dimensional incommensurability, isospectral ambiguity, high-frequency wrinkling, and spatial entanglement localization. Future research will focus on the complete machine-checked formalization of these theorems within interactive proof assistants such as LEAN 4, expanding the open verification corpus and establishing a new rigorous foundation for computational and physical multilinear geometry.

Invariant	Mathematical Family		Formal Definition	Physical / Computational Role
$\mathcal{W}_2(A, B)$	Optimal port	Trans-	$\inf_{\pi} \int \ x - y\ ^2 d\pi$	Geodesic distance between matrices of unequal dimensions ($n \neq m$).
$\kappa_{\text{BE}}(A)$	Metric Geometry	Measure	$\inf_{\xi} \langle (-\nabla^2 \log \Phi) \xi, \xi \rangle$	Ricci curvature, log-Sobolev constant, dissipation rate of flows.
$\text{Dgm}_k(A)$	Topological Analysis	Data	$\{(b_i, d_i) \mid H_k(X_t) \neq 0\}$	Multi-scale topological persistence of cavities; breaks isospectrality.
$E_{\text{pers}}(A)$	Persistent Entropy		$-\sum p_i \log p_i$	Quantifies topological disorder and complexity in entry landscapes.
$\mathcal{W}(\Phi(A))$	Geometric Measure Theory	Mea-	$\int_{\Sigma_t} H_t^2 d\mathcal{H}^{d-1} dt$	Total Willmore bending; penalizes second-order surface wrinkling.
$\mathcal{I}_A(v)$	GMT / Isoperimetry		$\inf \{ \text{Per}(E) \mid \mu_A(E) = v \}$	Cheeger constant, matrix bottlenecks, conductance bounds.
$\text{WF}(\Phi(A))$	Micro-Local Analysis		$\Sigma_{\text{sing}} \subset T^*\Omega \setminus \{0\}$	Exact spatial location and directional orientation of shock interfaces.
$s^*(A)$	Fractional Analysis	Analy-	$\sup \{s \mid [\Phi]_{B_{1,\infty}^s} < \infty\}$	Critical Besov exponent; detects fractal dimension of entries.
Tr_{ω}	Non-Commutative Geom.		$\text{Tr}_{\omega}(\Phi \mathcal{D} ^{-d})$	Conformal volume integral via Atiyah–Singer Dirac spectrum.
$\text{Vol}_{\text{QFI}}(\mathcal{T})$	Quantum Information		$\int_{\Omega} \sqrt{\det g^{\text{QFI}}(x)} d^d x$	Total distinguishability and parameter sensitivity of tensor states.
$\bar{x}_{\text{ent}}, \Sigma_{\text{ent}}^2$	Entanglement Contour		$\int x \mathcal{S}_{\text{cont}}(x) dx$	Physical center of mass and spatial spread of quantum entanglement.

TABLE 1. Comprehensive taxonomy of advanced geometric and topological invariants on functional realizations of matrices and tensors.

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